

1. Algorithm Design & Problem-Solving

1.1. Introduction

- **Abstraction:** filtering out and concentrating on the relevant information in a problem; allowing a programmer to deal with complexity
- **Decomposition:** breaking down problems into sub-problems in order to understand a process more clearly; program modules, procedures and functions all help the programmer to break down large problems
- **Algorithm:** a solution to a problem expressed as a sequence of steps

1.2. Identifier Table

- **Identifier:** name given to a variable in order to call it
- An identifier table depicts information about the variable, e.g.

Identifier	Data Type	Description
NumberDrawn	ARRAY[50] : BOOLEAN	FALSE – indicates the number has not yet been drawn TRUE – indicates the number has been drawn
Index	INTEGER	used as the subscript/index for the array

- **Rules for naming identifiers:**
 - Must be unique
 - Spaces must not be used
 - Must begin with a letter of the alphabet
 - Consist only of a mixture of letters and digits and the underscore character '_'
 - Must not be a 'reserved' word – e.g. Print, If, etc.

1.3. Basic Program Operations

- **Assignment:** an instruction in a program that places a value into a specified variable
- **Sequence:** programming statements are executed consequently, as they appear in the program
- **Selection:** control structure in which there is a test to decide if certain instructions are executed
 - **IF selection:** testing 2 possible outcomes
 - **CASE selection:** testing more than 2 outcomes
- **Repetition/Iteration:** control structure in which a group of statements is executed repeatedly
 - **FOR loop:** ++count-controlled++; executed a set no. of times
 - **WHILE loop:** ++pre-conditional++; executed based on condition at start of statements
 - **REPEAT loop:** ++post-conditional++; executed based on condition at end of statements

As for selecting what loop to use, it is best to use FOR loops when you know the number of iterations required, and a WHILE or REPEAT loop if you do not know the number of iterations required.

- Iterate over an array: FOR Loop
- Reading a file into a variable: WHILE Loop
- Asking for user input: WHILE/REPEAT Loop
- A loop that should execute n times: FOR Loop

1.4. Stepwise Refinement

- Process of developing a modular design by splitting a problem into smaller sub-tasks, which themselves are repeatedly split into even smaller sub-tasks until each is just one element of the final program.

1.5. Program Modules

- This refers to a modular program design
- **Subroutines:** self-contained section of code, performing a specific task; part of the main program
- **Procedures:** performs a specific task, no value returned to part of code where called
- **Functions:** performs a specific task, returns a value to part of code where called

1.6. Logic Statements

Operator	Meaning
<	Less than
<=	Less than/equal
>	Greater than
>=	Greater/equal
=	Equal to
<>	Not equal to